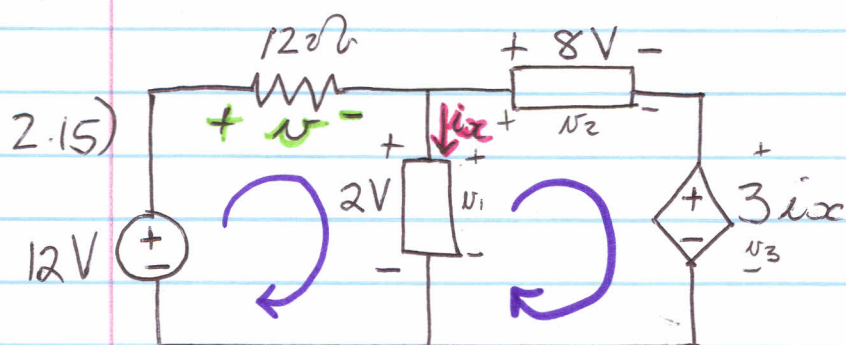


①

2.15; 2.16; 2.18; 2.20; 2.22; 2.25; 2.36; 2.38; 2.39;
2.42; 2.45

EBN 111: Tutoriaal 2

22 Feb 2013

Soek v ?
en i_x ?Vir v :**KVL** (Kirchhoff's voltage law)

$$\sum_{m=1}^M v_m = 0$$

(Verg 2.19, bl. 39)

$$+v + 2 - 12 = 0$$

$$v = 12 - 2$$

$$= 10 \text{ V}$$

Vir i_x :Gebruik **KVL** in tweede loop

$$\therefore \sum_{m=1}^M v_m = 0$$

$$\therefore -v_1 + v_2 + v_3 = 0$$

$$-2 + 8 + 3i_x = 0$$

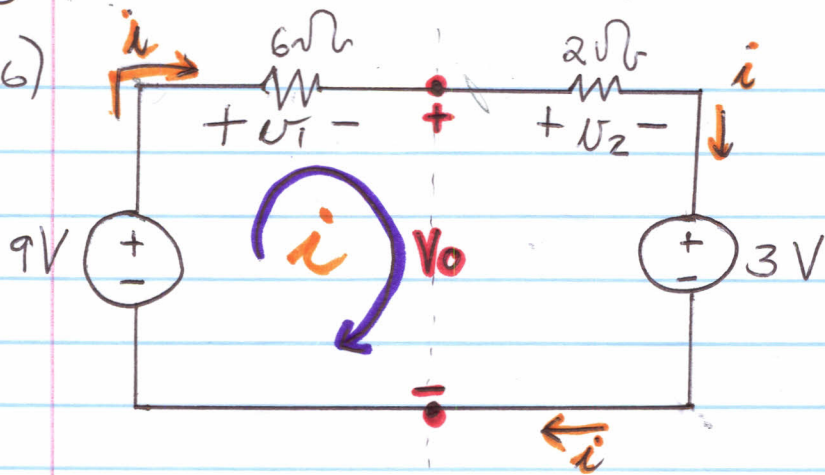
$$3i_x = -8 + 2$$

$$3i_x = -6$$

$$i_x = -2 \text{ A}$$

(2)

2.16)

Soek V_o ?

Stroom orals = i
 Hoekom?

Som van i voor
 node = Som van
 i na node. (B/38)

$$v = iR$$

KVL _{i} : $\sum_{m=1}^M v_m = 0$

$$\therefore -9 + v_1 + v_2 + 3 = 0$$

$$-9 + i(6) + i(2) + 3 = 0$$

$$8i = 6$$

$$i = \frac{6}{8} = \underline{0.75 \text{ A}}$$

Kry v_1 & v_2

$$\therefore v_1 = iR = 0.75 \times 6 \\ = 4.5 \text{ V}$$

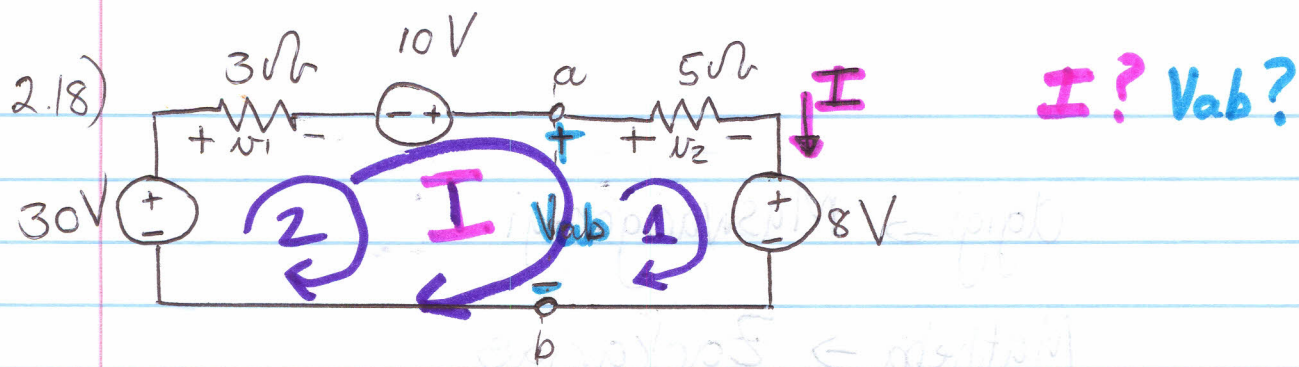
$$v_2 = iR = 0.75 \times 2 \\ = 1.5 \text{ V}$$

V_o :

$$-9 + v_1 + V_o = 0$$

$$V_o = 9 - v_1$$

$$\underline{V_o = 4.5 \text{ V}}$$



KVL:

$$\sum_{m=1}^M v_m = 0$$

$$\begin{aligned} -30 + v_1 - 10 + v_2 + 8 &= 0 \\ -30 + I(3) - 10 + I(5) + 8 &= 0 \\ 8I &= 32 \end{aligned}$$

$$\underline{I = 4 \text{ A}}$$

Soek v_1 & v_2 :

$$v_1 = IR = (4)(3) = 12 \text{ V}$$

$$v_2 = IR = (4)(5) = 20 \text{ V}$$

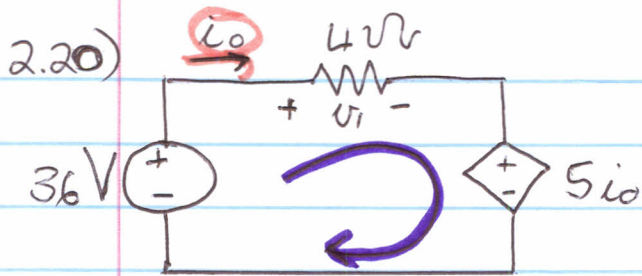
$$-V_{ab} + v_2 + 8 = 0$$

$$V_{ab} = v_2 + 8$$

$$\underline{V_{ab} = 28 \text{ V}}$$

Toets linkerkantse stroombaan:

$$\begin{aligned} V_{ab} - 30 + v_1 - 10 &\neq 0 \\ &= 28 - 30 + 12 - 10 \\ &= 0 \end{aligned}$$

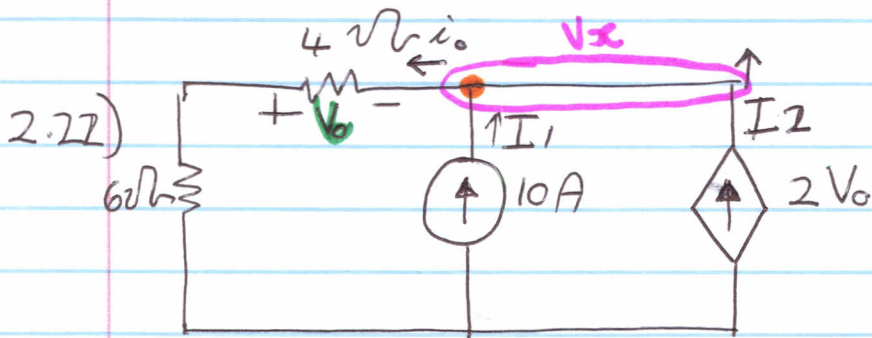


Soek i_o ?

KVL: $-36 + v_1 + 5i_o = 0$
 ($v_1 = i_o(4)$): $-36 + 4i_o + 5i_o = 0$

$$9i_o = 36$$

$$\underline{i_o = 4 \text{ A}}$$



a) V_o ?

KCL: Som van i by node is nul

$$\sum_{n=1}^N i_n = 0$$

(Verg 2.13, b/37)

$$\begin{aligned} \therefore I_1 + I_2 - i_o &= 0 \\ i_o &= I_1 + I_2 \\ i_o &= 10 + 2V_o \end{aligned}$$

b) Drywing geleier?

$$\begin{aligned} p &= i^2 R \quad (2 \cdot 10) \\ p &= iV \end{aligned}$$

$$\begin{aligned} \therefore V_o &= iR \\ -V_o &= i_o(4) \\ V_o &= -4i_o \\ V_o &= -4(10 + 2V_o) \\ V_o &= -40 - 8V_o \\ 9V_o &= -40 \\ V_o &= -4.44 \text{ V} \end{aligned}$$

$$V_o = \frac{-4}{6+4} \cdot V_x \quad (\text{Voltage division})$$

$$V_o = -2.5V_x$$

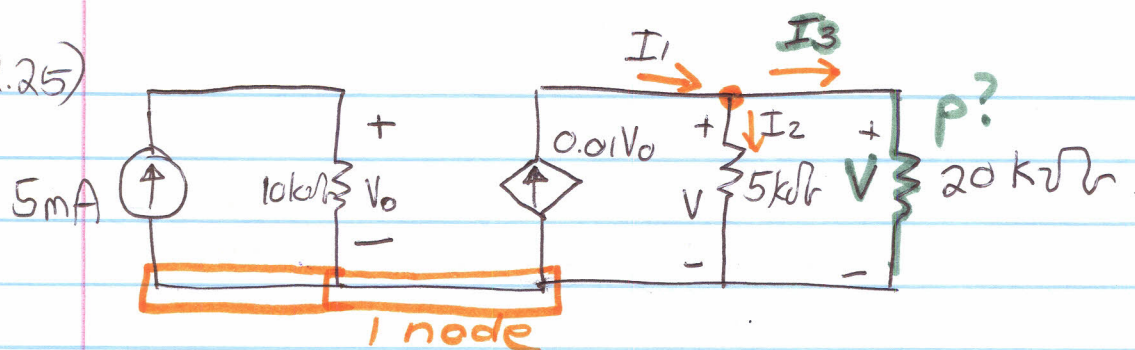
$$\therefore V_x = \frac{-4.44}{-2.5}$$

$$V_x = 1.776$$

b). Power

$$\begin{aligned} p &= V_x I_2 \\ p &= (1.776)(2V_o) \\ p &= -15.77 \text{ W} \end{aligned}$$

2.25)



Soek V?
Is?
Pack?

Vir V_o : $V_o = iR$ (Verg 2.3)
 $V_o = (5 \times 10^{-3})(10 \times 10^3)$
 $V_o = 50 \text{ V}$

$$I_1 = (0.01)(50) = 0.5 \text{ A}$$

$$U = \mathbb{R}.$$

KCL:

$$I_1 = I_2 + I_3$$

$$0.5 = \frac{V}{5 \times 10^3} + \frac{V}{20 \times 10^3}$$

$$V \left(\frac{1}{5 \times 10^3} + \frac{1}{20 \times 10^3} \right) = 0.5$$

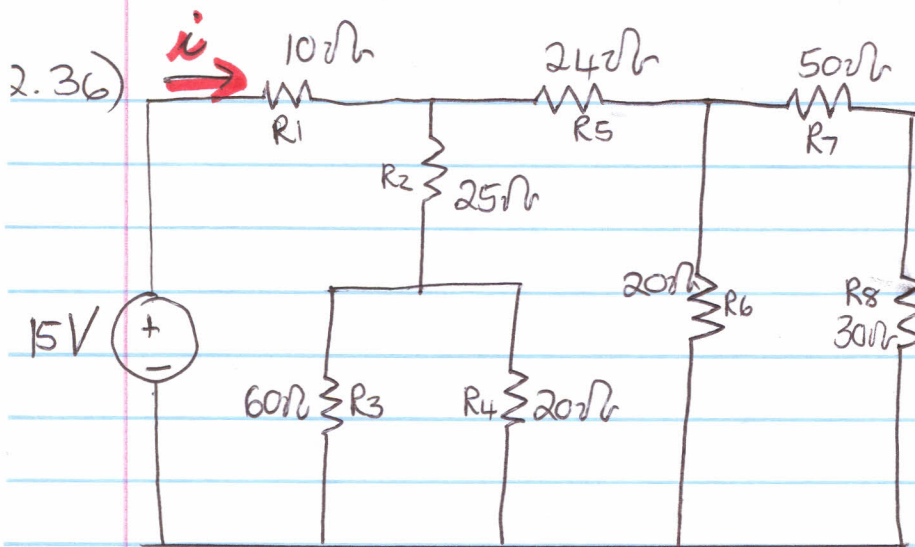
$$V = 2000 \text{ (V)}$$

$$I_3 = \frac{2000}{20 \times 10^3} = 0.1 \text{ A}$$

$$I_2 = \frac{2000}{5 \times 10^3} = 0.4 \text{ A.}$$

$$\therefore I_1 = 0.1 + 0.4 = 0.5 \text{ A}$$

$$P_{\text{rok}} = (0.1)(2000) = \underline{200 \text{ W}}$$



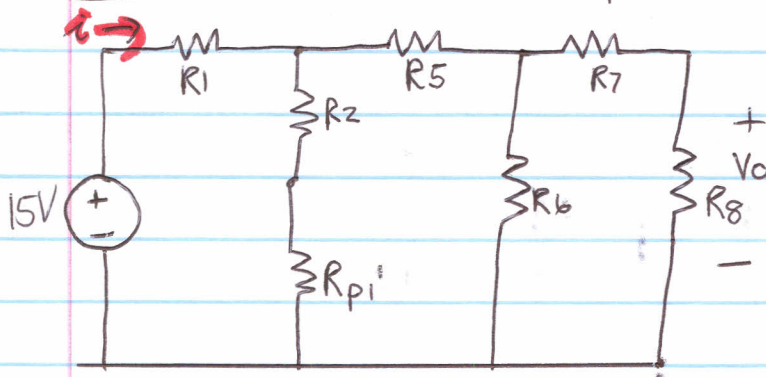
Soek i & V_o ?



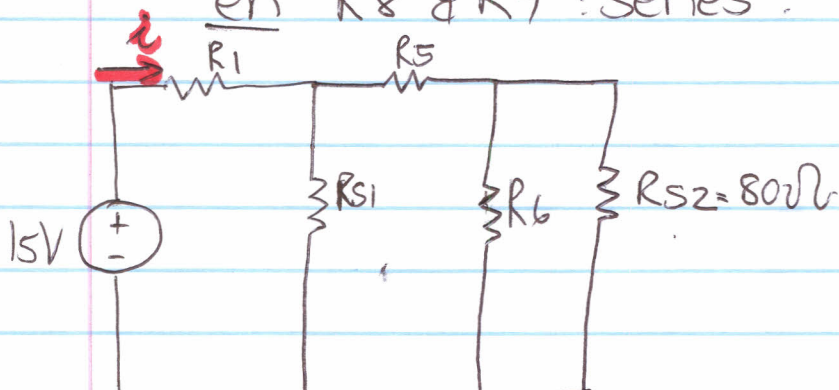
Stap 1: R_3 & R_4 parallel = $R_{p1} = \frac{R_3 R_4}{R_3 + R_4}$

$$R_{p1} = \frac{(60)(20)}{60 + 20}$$

$$R_{p1} = 15\Omega$$

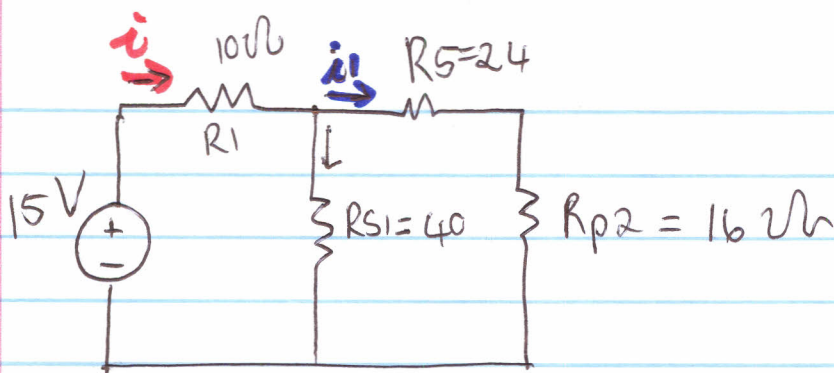


Stap 2: R_2 & R_{p1} series: $R_{s1} = R_2 + R_{p1} = 40\Omega$
 en R_8 & R_7 series: $R_{s2} = R_8 + R_7 = 80\Omega$

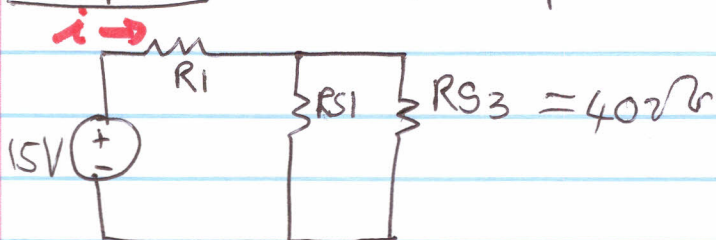


Stap 3: $R_{s2} \parallel R_6$ $R_{p2} = \frac{(R_{s2})(R_6)}{R_{s2} + R_6} = 16\Omega$



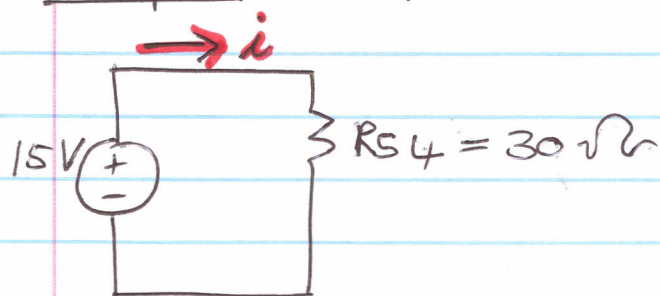


Step 4: $\Downarrow$ R_2 in series $R_5: R_3 = 40\Omega$



Step 5: $R_{S1} \parallel R_{S3} \therefore R_{p3} = \frac{(R_{S1})(R_{S3})}{R_{S1} + R_{S3}}$
 $\Downarrow$ $R_{p3} = 20\Omega$

Step 6: R_{p3} in Series $R_1 \therefore R_{S4} = 20 + 10$
 $R_{S4} = 30\Omega$



$$\therefore i = \frac{V}{R} = \frac{15}{30}$$

$$\underline{i = 0.5A}$$

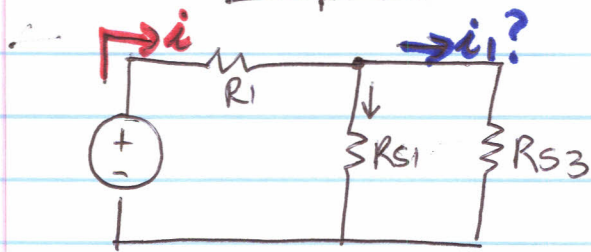
Determine V_o

Terug na Step 1: Back to step 1



mark i_1
 & i_2

① First determine i_1 :
use Step 4



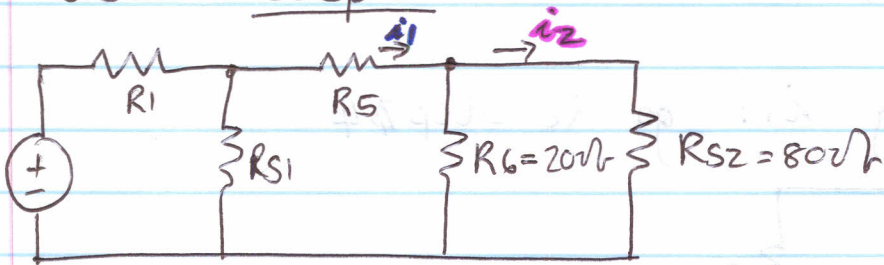
— use current division

$$i_1 = \left(\frac{R_{51}}{R_{51} + R_{53}} \right) i \quad (\text{equ 2.43})$$

$$\text{Thus } i_1 = \left(\frac{40}{40 + 40} \right) i \quad (i = 0.5 \text{ A})$$

$$i_1 = 0.25 \text{ A}$$

② Determine i_2
use step 2:



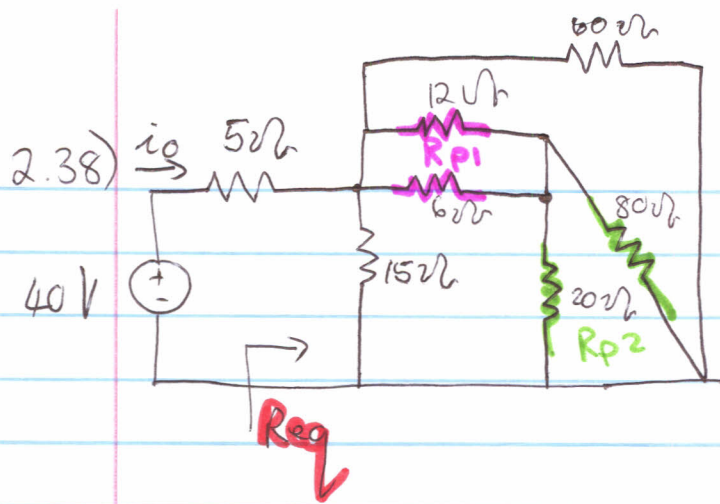
$$\text{use current division: } i_2 = \left(\frac{R_6}{R_6 + R_{52}} \right) i_1$$

$$i_2 = \left(\frac{20}{20 + 80} \right) 0.25$$

$$i_2 = 0.05 \text{ A}$$

For V_o

$$\begin{aligned} \therefore V_o &= i_2 R_8 \\ &= (0.05)(30) \\ &= 1.5 \text{ V} \end{aligned}$$

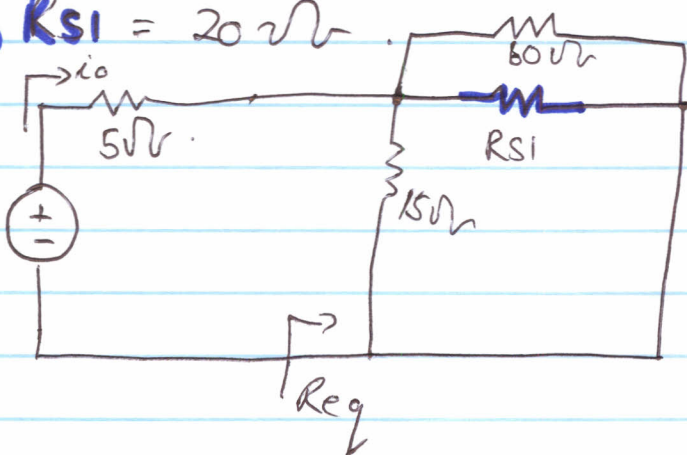


$R_{eq}?$

series $R_{p1} = \frac{(12)(6)}{12+6} = 4\Omega$

$R_{p2} = \frac{(80)(20)}{80+20} = 16\Omega$

$R_{s1} = 20\Omega$



$$R_{p3} = (R_{s1}) \parallel 60\Omega$$

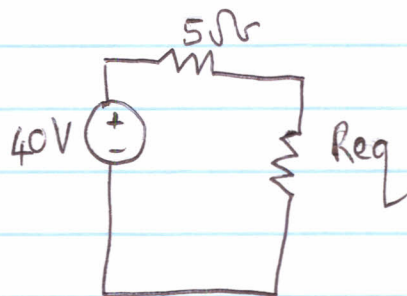
$$= \frac{(20)(60)}{60+20}$$

$$= 15\Omega$$

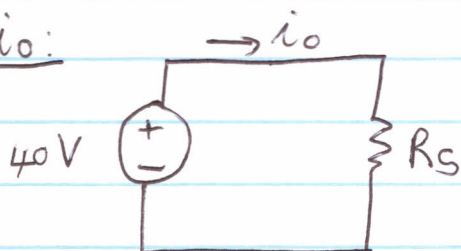
$$R_{eq} = 15\Omega \parallel R_{p3}$$

$$= \frac{(15)(15)}{15+15}$$

$R_{eq} = 7.5\Omega$



Vir i_o :



$$R_s = 5\Omega + R_{eq}$$

$$R_s = 12.5\Omega$$

$$i = \frac{V}{R_s} = \frac{40}{12.5}$$

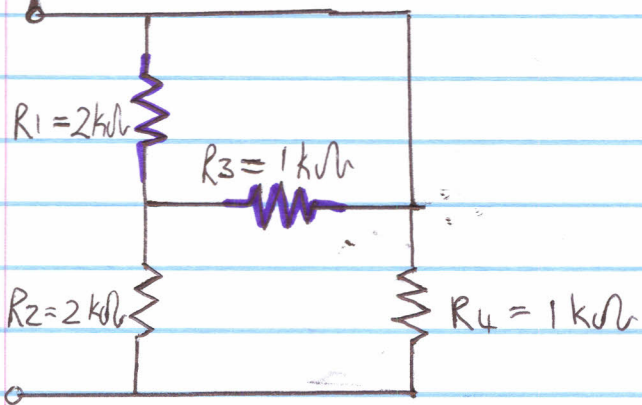
$i_o = 3.2\text{ A}$

2.39)

Req?

a)

Req



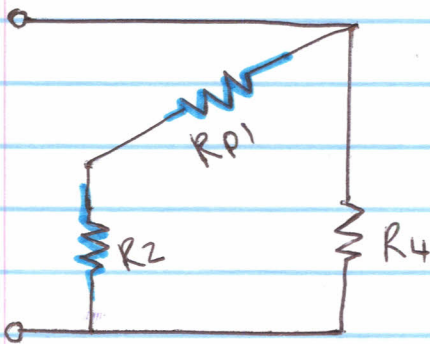
$$R_{p1} = R_1 \parallel R_3$$

$$= \frac{(2000)(1000)}{2000 + 1000}$$

$$= \frac{2000}{3} \Omega$$



Req



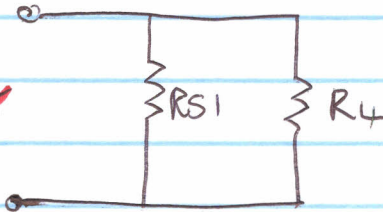
$$R_{s1} = R_{p1} + R_2$$

$$= \frac{2000}{3} + 2000$$

$$R_{s1} = \frac{8000}{3}$$



Req

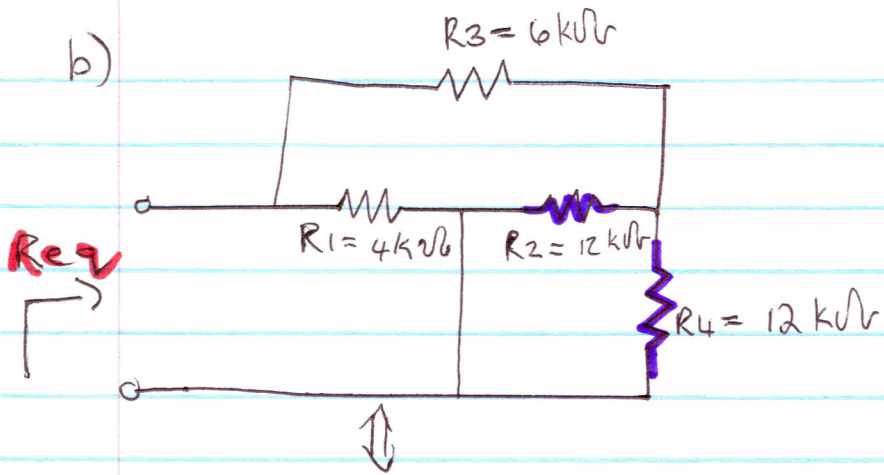


$$\therefore R_{eq} = R_{s1} \parallel R_4$$

$$= \frac{\left(\frac{8000}{3}\right)(1000)}{\frac{8000}{3} + 1000}$$

$$\underline{R_{eq} = 727.27 \Omega}$$

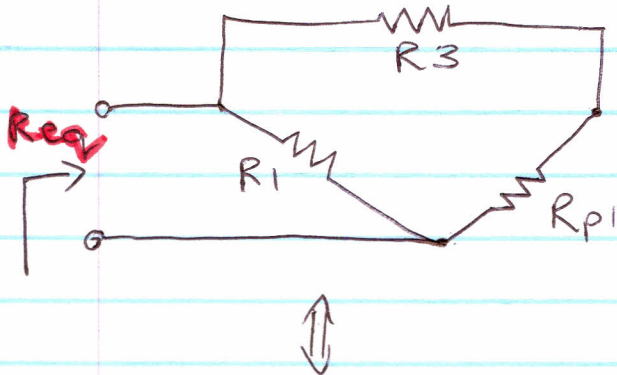
b)



$$R_{p1} = R_2 \parallel R_4$$

$$= \frac{(12k)(12k)}{12k + 12k}$$

$$R_{p1} = 6k$$

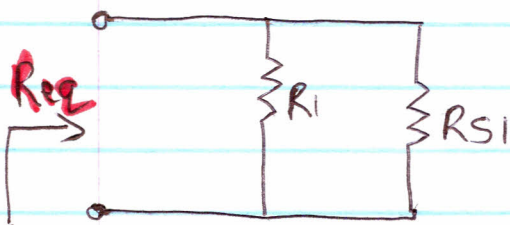


R_{p1} in series R_3

$$\therefore R_{s1} = R_3 + R_{p1}$$

$$= 6k + 6k$$

$$= 12k\Omega$$



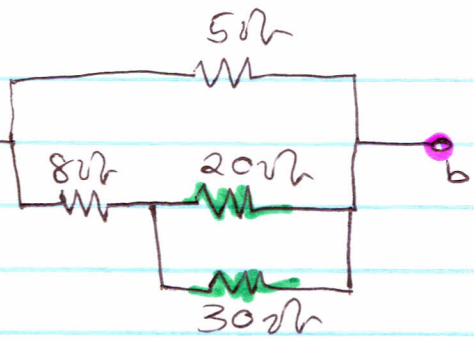
$$\therefore R_{eq} = R_1 \parallel R_{s1}$$

$$= \frac{(4k)(12k)}{4k + 12k}$$

$$\underline{R_{eq} = 3k\Omega}$$

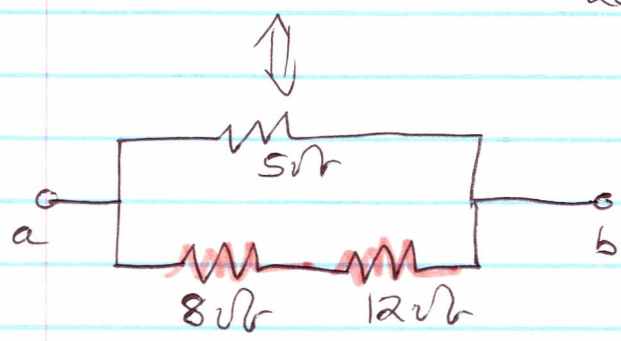
2.42)

a)

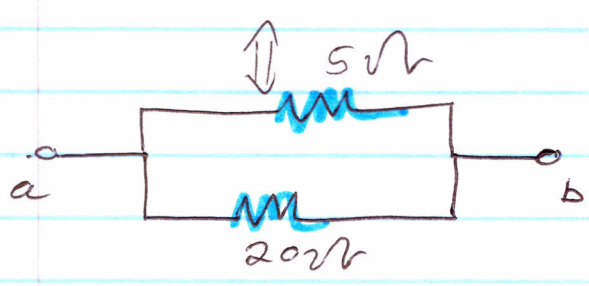


Single  between a and b

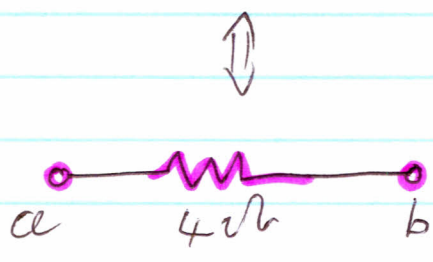
Step 1: $20 \parallel 30 = \frac{(20)(30)}{20 + 30} = 12 \Omega$

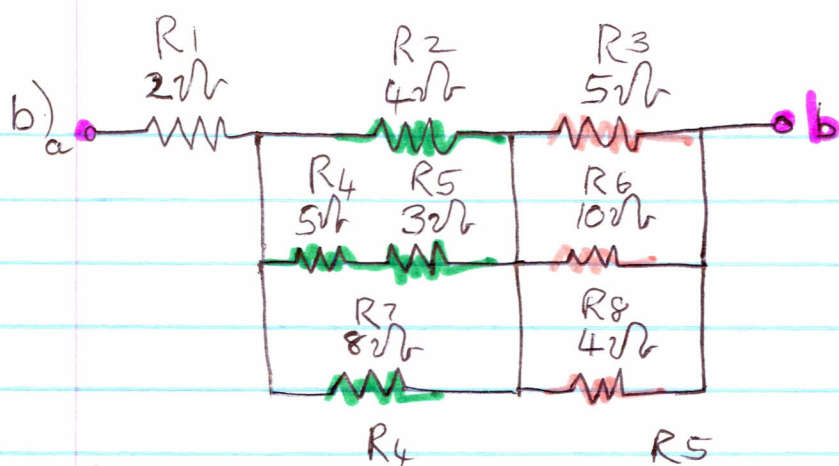


Step 2: 8Ω in serie $12 \Omega = 20 \Omega$



Step 3: $20 \Omega \parallel 5 \Omega = \frac{(20)(5)}{20 + 5} = 4 \Omega$





Step 1: a) 5Ω in series $3\Omega = 8\Omega = R_{S1}$

b) $R_2 \parallel R_{S1} \parallel R_7$ $\therefore \frac{1}{R_{P1}} = \frac{1}{R_2} + \frac{1}{R_{S1}} + \frac{1}{R_7}$

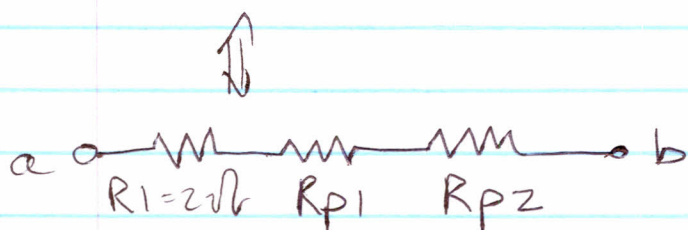
$$\frac{1}{R_{P1}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$

$$R_{P1} = 2\Omega$$

c) $R_8 \parallel R_6 \parallel R_3$ $\therefore \frac{1}{R_{P2}} = \frac{1}{R_8} + \frac{1}{R_6} + \frac{1}{R_3}$

$$\frac{1}{R_{P2}} = \frac{1}{4} + \frac{1}{10} + \frac{1}{5}$$

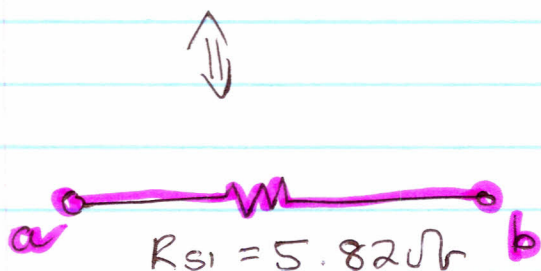
$$R_{P2} = \frac{20}{11}\Omega$$



$$R_{S1} = R_1 + R_{P1} + R_{P2}$$

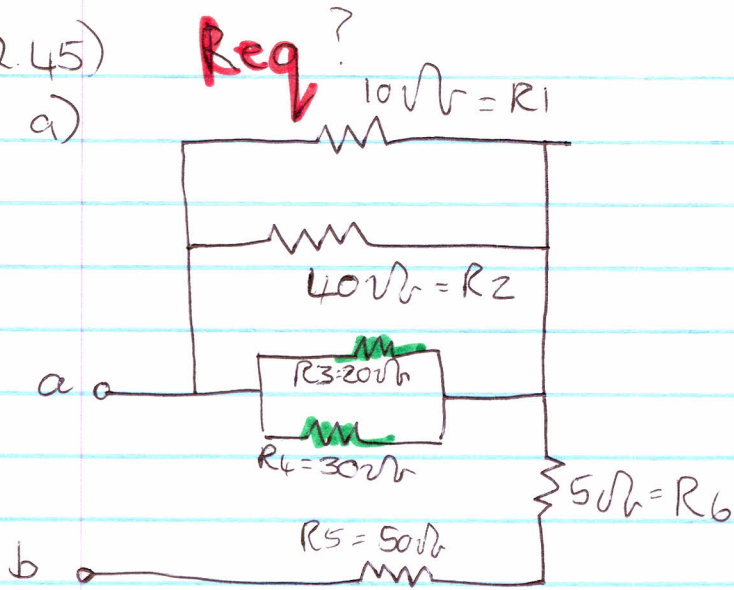
$$R_{S1} = 2 + 2 + \frac{20}{11}$$

$$R_{S1} = 5.82\Omega$$



2.45)

a)



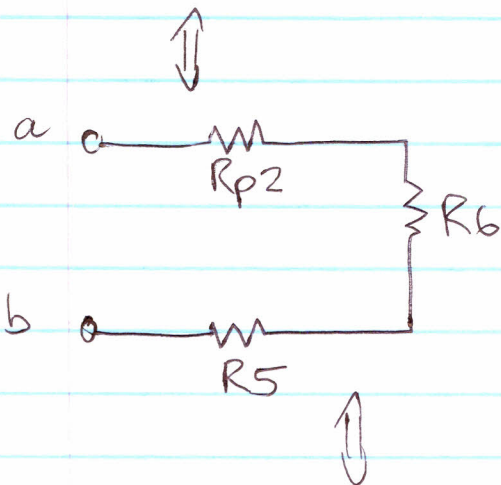
Step 1: a) $R_3 \parallel R_4$ $\therefore R_{p1} = \frac{(R_3)(R_4)}{R_3 + R_4} = \frac{(20)(30)}{20 + 30} = 12\Omega$

b) $R_{p1} \parallel R_2 \parallel R_1$

$$\frac{1}{R_{p2}} = \frac{1}{R_{p1}} + \frac{1}{R_2} + \frac{1}{R_1}$$

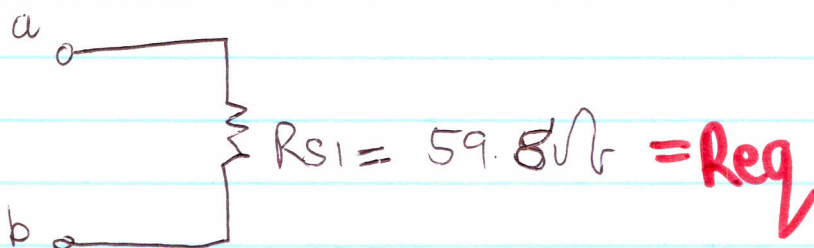
$$\frac{1}{R_{p2}} = \frac{1}{12} + \frac{1}{40} + \frac{1}{10}$$

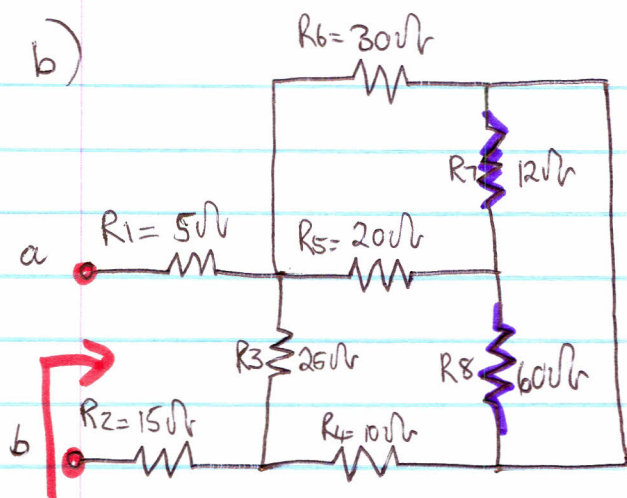
$$R_{p2} = 4.8\Omega$$



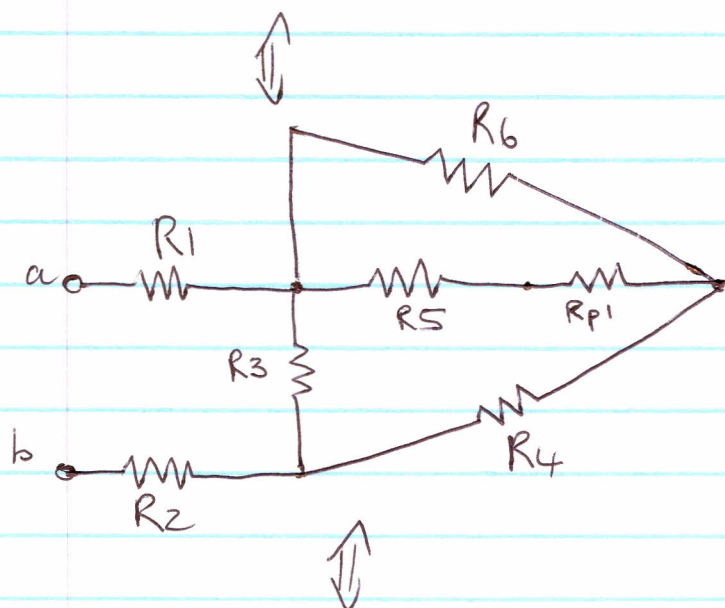
$$R_{S1} = R_{p2} + R_6 + R_5$$

$$= 59.8\Omega \rightarrow$$



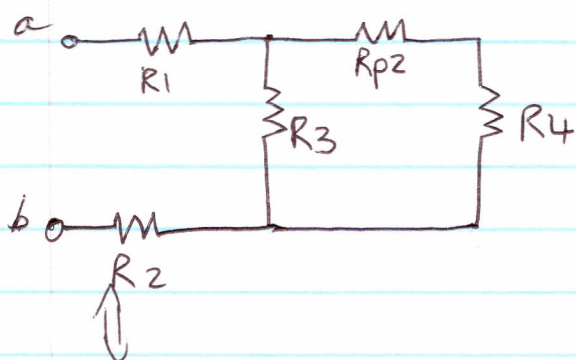


Step 1: $R_7 \parallel R_8 \therefore R_{p1} = \frac{(R_7)(R_8)}{R_7 + R_8} = \frac{(12)(60)}{12 + 60} = \underline{10\Omega}$



Step 2: a) R_5 in series R_{p1}
 $R_{s1} = R_5 + R_{p1} = 30\Omega$

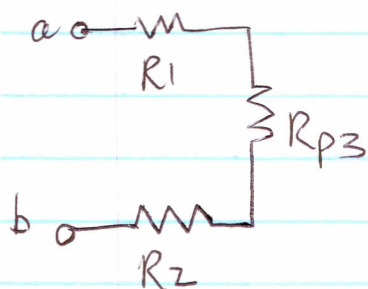
b) $R_{s1} \parallel R_6 \therefore R_{p2} = \frac{(R_6)(R_{s1})}{R_6 + R_{s1}} = \frac{(30)(30)}{30 + 30} = 15\Omega$



Step 3: a) $R_{s2} = R_{p2} + R_4$
 $R_{s2} = 25\Omega$

b) $R_{s2} \parallel R_3 \therefore R_{p3} = \frac{(R_3)(R_{s2})}{R_3 + R_{s2}} = 12.5\Omega$

Step 4: $R_1, R_{p3} \& R_2$ in series



$R_{eq} = R_1 + R_{p3} + R_2 = 32.5\Omega$